Student's Name: ……

Arusha Regional Annual Examination December 2022

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM THREE ANNUAL EXAMINATION
BIOLOGY

Time: 3:00 Hours **01st DECEMBER 2022**

Instructions

1. This paper consists three sections; A, B and C with a total of fifteen (15) questions.
2. Answer **all** questions in sections A, B, and **two (2)** questions from section C of which 13 is compulsory.
4. All writing should be in blue ink, except for diagrams, which must be drawn in pencil.
6. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Name or Examination Number** on every page of your answer sheets.

SECTION A (15 Marks)

Answer all questions

1. Choose the most correct answer among the given alternatives and write down the answer in the space given.

(i) The Monera are also referred to as Prokaryotes. This means having:

A. No nucleus B. Membrane bounded organelles C. Simple structure
D. Reduced nucleus. E. Circular nucleus.

(ii) Biology experiments are used to test:

A. Facts. B. Hypothesis. C. Laws D. Problems. E. Ability of species to survive.

(iii) One of the following is the importance of excretion.

A. It provides useful substances such as latex, which is used to make rubber
B. It used in learning Biology C. It provides suitable working environment for the cells.
D. It helps in preventing hormonal imbalance problems
E. It is the process that leads to energy production in the body

(iv) Select the item that indicates the best match in vitamin deficiency:

A. Vitamin A …… dry scaly skin B. Vitamin B …… loss of appetite and yawning
C. Vitamin C …… anemia and high blood pressure D. Vitamin D …… soft deformed bones
E. Vitamin K …… poor night vision

(v) Which of the following shows the passage of water through the cells and tissues when it enters a plant?

A. Root hair →spongy mesophyll → xylem B. Root hair → phloem → spongy mesophyll.
C. Root hair → palisade mesophyll → xylem. D. Spongy mesophyll → xylem → root hair
E. Root hair → xylem → spongy mesophyll.

(vi) When the heart of mammals contracts, blood in the right vertical is pumped out

A. Through the pulmonary artery. B. Through the pulmonary vein. C. Into the right auricle.
D. Through the aorta. E. Into the coronary arteries.

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(vii) Which of the following is not a sexually transmitted disease?
A. Trichomoniasis B. Candidiasis C. Vaginosis D. trypanosomiasis E. AIDS

(viii) On average, 90% of energy is lost at each energy transfer in a food chain. Which of the following is the major cause of this energy loss?
A. Growth B. Cell repair C. Digested material D. Movement E. Respiration

(ix) The coiled tube in which mature sperms are stored is known as:
A. Penis B. Scrotum C. Vas deferens D. Epididymis E. Testes

(x) Which of the following is common to both plants and animals?
A. Respiration B. Digestion C. An excretory system D. Chloroplasts E. Starch grains

2. Match items in list A with the correct phrase in list B.

LIST A	**LIST B**
(i) Cell sensitive to coloured light intensity	A. Cerebellum
(ii) Promotes conversion of glycogen into glucose to be used in the body.	B. Rods
(iii) Controls and coordinates body balancing organs, thus making accurate movement possible.	C. Glucagon
(iv) Corrects short sight	D. Thyroid gland
(v) Produces hormone that facilitates the uptake of calcium and phosphate by bones.	E. Cones
	F. Diverging lens
	G. Converging lens
	H. Cerebrum

SECTION B (60 Marks)
Answer all questions

3. Spoilage of food leads to food loss and consequently food shortage in general. Advice six measures stakeholders could take in order to significantly reduce the problem.
4. (a) State one example of a population of organisms and describe any two competitive interactions that can be observed in that population.
   (b) Briefly explain why there are fewer predators than prey.
5. Cross pollination enables mixing of genetic qualities between different plants, hence hybrid vigor is obtained. Explain four adaptations plants have developed to exploit this advantage.
6. (a) (i) Briefly explain how the vascular bundles arrangement in the stem of a mango tree differ from the arrangement of the vascular bundles in in the stem of a coconut tree.
   (ii) Briefly explain how transport of water, mineral salts and nutrients in a moss plant differ from the transport of the same materials in a maize plant.
   (b) State three adaptations of the plants' roots for absorption of water and mineral salts from the soil.
7. (a) In at least two points, briefly explain the significance of sexual reproduction.
   (b) Briefly explain four ways in which asexual reproduction is more advantageous when compared to sexual reproduction.
8. After running fast during a school running competition, a student began to breathe very fast and experienced sore muscles. After resting for some time the soreness in the muscles was gone and his breathing went back to normal.
   (a) Briefly explain how the soreness in the muscles developed.
   (b) Briefly explain the relationship between breathing fast, resting, and disappearance of the sore muscles condition.

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9. John complains of not seeing near objects clearly while Jane can see near objects clearly but cannot see distant objects as clearly as John can. Describe the conditions experienced by John and Jane, and illustrate the corrective measures that could be taken.
10. (a) Briefly explain the three principles of waste disposals.
    (b) Briefly educate people about any three ways poor disposal of waste can negatively affect them.
11. (a) Briefly explain the difference between regulation and homoeostasis
    (b) Giving at least two points, explain the importance of regulation in the survival of organisms. (c) Briefly explain the mechanism of osmoregulation.
12. (a) Mr. Seypalm injects himself with doses of insulin every time before taking his meal.
    (i) What is he trying to control with the insulin shots?
    (ii) What does the insulin do to keep whatever he is trying to control in check?
    (iii) What is the name of the medical condition he might be experiencing?
    (iv) State two reasons that could be the physiological sources of his medical condition.
    (b) How each of the following plant is adapted survive in its habitat?
    (i) Halophytes
    (ii) Mesophytes
    (iii) Xerophytes

SECTION C (25 Marks)

Answer two questions from this section: Question 13 is compulsory

13. Figure 1 shown below is an organism belonging to the plant kingdom.



Figure 1

Write an essay about the division to which the plant in figure 1 belong, explaining the characteristics that make the division distinct from other plants divisions, and the economic importance of the plants belonging the division. (15 marks)

14. Explain transpiration in plants using the following guidelines:
    - Meaning of transpiration
    - The importance of transpiration
    - The adaptation features which enable plants to reduce water loss (10 marks)
15. Mineral ions are an important component of a healthy diet. Describe how the other components of a healthy diet are important in keeping us healthy. (10 marks)